Design Studio Lighting Audit

Survey Date: 2026-08-14 15:55 | **Surveyor/Author:** Lighting Engineer | **Room:** Main Design Studio

Room Dimensions:	7.00 m (L) × 5.00 m (W)	**Floor Area:**	35.00 m²
Ceiling Height:	3.00 m	**Working Plane Height:**	0.75 m
Wall Margin Band:	0.50 m	**Grid Points Total:**	20 points (20 measured)

Photometric Numerical Analysis

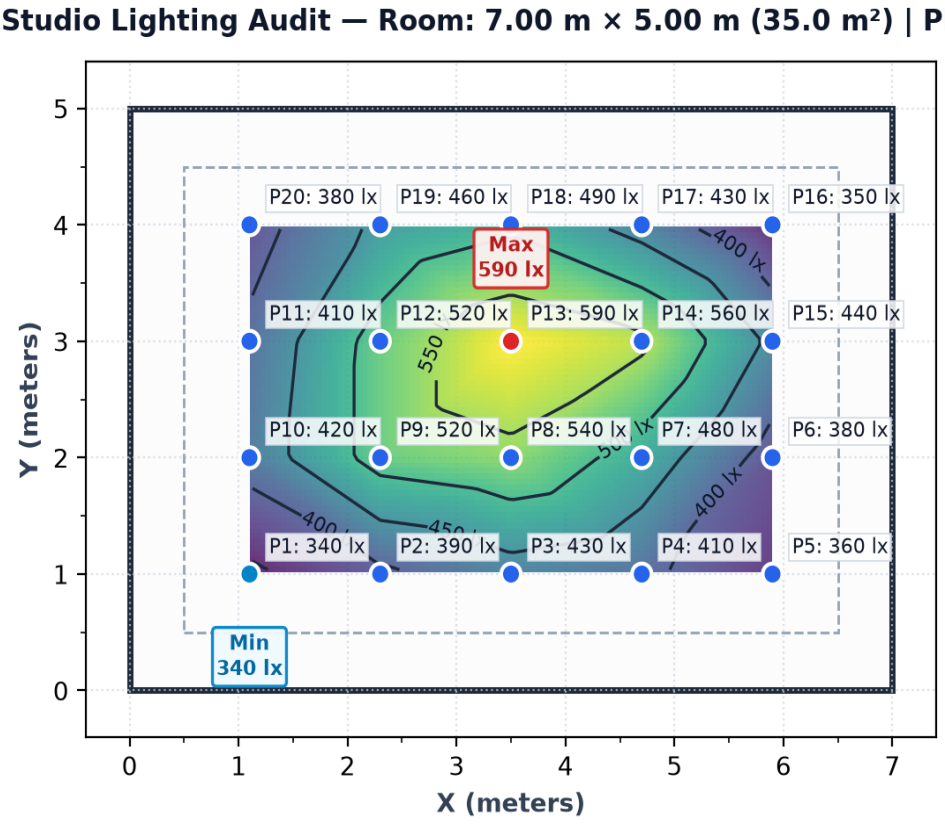
Average Illuminance (E_{avg})	**445.0 lx**	**Uniformity U_0 (E_{min} / E_{avg})**	**0.764**
Minimum Illuminance (E_{min})	340.0 lx at P1 (1.10, 1.00) m	**Uniformity U_1 (E_{min} / E_{max})**	0.576
Maximum Illuminance (E_{max})	590.0 lx at P13 (3.50, 3.00) m	**Standard Deviation (σ)**	72.7 lx
Range (E_{max} – E_{min})	250.0 lx	**Median Illuminance**	430.0 lx

Comparison with Selected Target Illuminance

Target Reference: 500 lx | **Points Evaluated:** 20 | **Points Below Target:** 15 (75.0%) | **Points ≥ Target:** 5 (25.0%) | **Mean Abs Deviation:** 78.0 lx

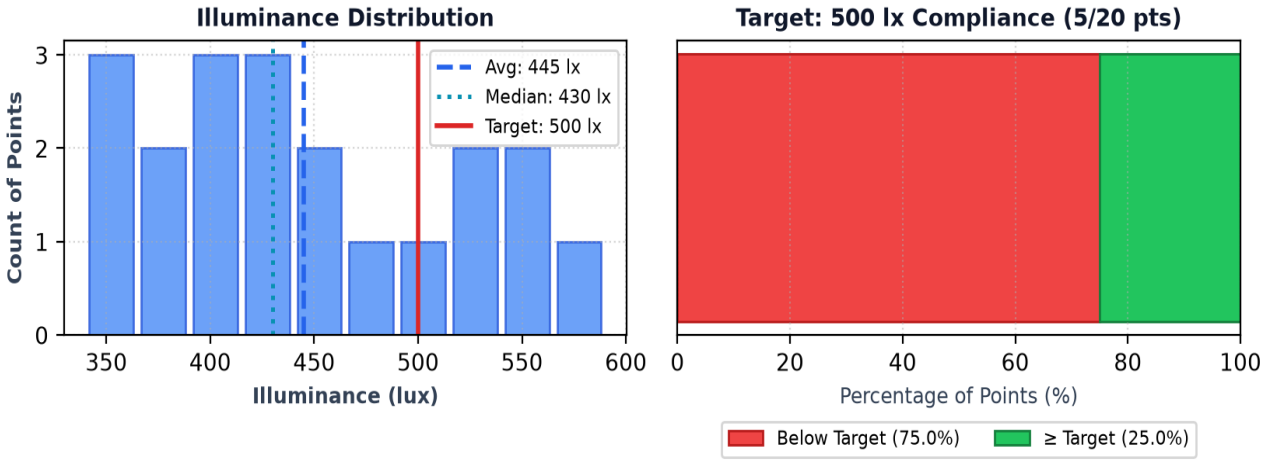
Notice: Target entered by user: 500 lx. 15 of 20 measurement points are below the selected target (75.0%).

2D Illuminance Distribution Map & Isolux Contours



Scientific Methodology Note: *The continuous false-colour map and isolux contour lines represent numerical spatial estimates reconstructed between physical measurement locations. Physical measurement markers (points) represent actual readings recorded at the working plane with a physical lux meter. Interpolated values between points are mathematical estimates and are not presented as direct measurements.*

Illuminance Frequency Distribution & Compliance



Measurement Schedule & Location Coordinates

Point ID	X (m)	Y (m)	Measured Lux	Expected Lux	Δ E (lx)	Notes / Instrument
P1	1.10	1.00	340.0	360.0	-20.0	Grid (1, 1)
P2	2.30	1.00	390.0	400.0	-10.0	Grid (2, 1)
P3	3.50	1.00	430.0	450.0	-20.0	Grid (3, 1)
P4	4.70	1.00	410.0	430.0	-20.0	Grid (4, 1)
P5	5.90	1.00	360.0	380.0	-20.0	Grid (5, 1)
P6	5.90	2.00	380.0	400.0	-20.0	Grid (5, 2)
P7	4.70	2.00	480.0	500.0	-20.0	Grid (4, 2)
P8	3.50	2.00	540.0	550.0	-10.0	Grid (3, 2)
P9	2.30	2.00	520.0	530.0	-10.0	Grid (2, 2)
P10	1.10	2.00	420.0	440.0	-20.0	Grid (1, 2)
P11	1.10	3.00	410.0	430.0	-20.0	Grid (1, 3)
P12	2.30	3.00	520.0	530.0	-10.0	Grid (2, 3)
P13	3.50	3.00	590.0	600.0	-10.0	Grid (3, 3)
P14	4.70	3.00	560.0	570.0	-10.0	Grid (4, 3)
P15	5.90	3.00	440.0	450.0	-10.0	Grid (5, 3)
P16	5.90	4.00	350.0	370.0	-20.0	Grid (5, 4)
P17	4.70	4.00	430.0	440.0	-10.0	Grid (4, 4)
P18	3.50	4.00	490.0	500.0	-10.0	Grid (3, 4)
P19	2.30	4.00	460.0	470.0	-10.0	Grid (2, 4)
P20	1.10	4.00	380.0	390.0	-10.0	Grid (1, 4)